

Data Elements from Raw Data: A KAN-T-GCN Model for Traffic Speed Prediction

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Abstract: This paper explores the development of interpretable data elements from raw data using Kolmogorov-Arnold Networks (KAN). With the exponential growth of data in contemporary society, there is an urgent need for effective data processing methods to unlock the full potential of this resource. The study focuses on the application of KAN in the transportation sector to transform raw traffic data into meaningful data elements. The core of the research is the KAN-T-GCN model, which synergizes Kolmogorov-Arnold Networks with Temporal Graph Convolutional Networks (T-GCN). This innovative model demonstrates superior performance in predicting traffic speeds, outperforming existing methods in terms of accuracy, reliability, and interpretability. The model was evaluated using real-world datasets from Shenzhen, Los Angeles, and the San Francisco Bay Area, showing significant improvements in different metrics. The paper highlights the potential of KAN-T-GCN to revolutionize data-driven decision-making in traffic management and other sectors, underscoring its ability to handle dynamic updates and maintain data integrity.

Keywords: data element; KAN-T-GCNs; KANs; traffic speed forecasting

I. INTRODUCTION

Data has become indispensable across all industries in contemporary society, with vast quantities generated daily or even hourly due to rapid industrial development. While some data is reinvested into industry growth, a significant portion remains stored due to inadequate transmission media for data sharing, limiting its reusability [1]. Consequently, the full potential of data has yet to be realized. To address this, the concept of data elements has been proposed. Data, akin to other production factors, serve as primary drivers of social development. They are comparable to raw materials or essential resources in traditional industries, playing a crucial role in the operation and progression of modern economies [2].

Traditional production factors—land, labor, capital, and technology—have evolved from “resources” to “factors” and then to “circulation” throughout history [3]. This evolution has been accompanied by continuous market system improvements and optimal resource allocation. Through multiple rounds of resource allocation and market mechanism evolution, these four major production factors have gradually formed a stable and complete market system. By summarizing the marketization process of these production factors, it becomes evident that they follow a three-tier market structure development model. Initially, raw resources undergo a standardization process to form an “intermediate state.” This intermediate state acts as a bridge connecting raw resources and fi-

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nal products, possessing clear transactional attributes and value measurement standards, making it a widely recognized and accepted market entity. Through this standardized intermediate state, various production factors can circulate freely in the market, achieving optimal allocation and efficient resource utilization. For instance, land resources are developed and prepared to become usable land for real estate or factory construction; labor resources acquire work certificates through education and training to participate in production labor [4]. Similarly, data elements, which are basic units of information with unique meanings and distinct subcategories of value, can act as the intermediate state in the data industry [5]. After systematic organization, in-depth analysis, and professional training, raw data sets are transformed into applicable tools for specific fields. These tools not only promote the development of the original data set's source field but also provide practical application value and reference for other similar fields. For example, in the transportation sector, integrating key vehicle data such as network freight can construct high-quality dynamic data sets covering vehicle operational behavior and accident statistics. These data sets support differentiated credit, insurance services, second-hand car consumption, and more. This further promotes the construction and reuse of high-quality transportation data sets, fosters the development of industry-specific artificial intelligence platforms and tools, and helps enterprises improve transportation efficiency [6].

Given the central role of data elements in the data industry [7], several key characteristics are essential: precision, dynamic update capability, and interpretability. As an intermediate form of data, data elements should accurately extract the most valuable information from data sets, such as overall data trends and future predictions. With the continuous development of the industry, original data sets used to generate data elements will persistently incorporate new data. Therefore, data elements must have the ability to update dynamically according to the original data sets to adapt to these changes. It is crucial to minimize resource consumption required for these updates. Additionally, despite incorporating new data, data elements should maintain the information carried by old data to ensure continuity and integrity. Interpretability is an indispensable attribute of data elements, requiring models to evaluate their value and support human

interaction and practical scientific exploration [8].

This paper selects relevant data from the transportation field as examples of the original data set and uses Kolmogorov-Arnold Networks [9] to generate data elements. Traffic prediction is increasingly vital to society as it directly influences urban planning, public safety, and the efficiency of transportation systems. Accurate traffic prediction models, such as those reviewed by [10], enable real-time adjustments to traffic flow, thereby reducing congestion, lowering the risk of accidents, and improving overall road safety. Beyond these benefits, traffic prediction has far-reaching practical implications. It plays a pivotal role in dynamic urban mobility systems by enabling adaptive signal controls and optimized route planning, directly improving travel times and fuel efficiency. In logistics, accurate forecasts enhance delivery scheduling and reduce operational costs. Moreover, in emergency response scenarios, real-time traffic predictions ensure faster routing of ambulances and fire trucks, which can be life-saving. The data elements extracted by Kolmogorov-Arnold Networks exhibit high accuracy, ensure good interpretability, and support subsequent dynamic updates.

In addition to its applications in transportation, the methodology proposed in this study has potential relevance across various fields where spatiotemporal predictions are critical. For instance, in healthcare, similar models have been applied to predict patient flow in emergency departments and optimize resource allocation, improving patient outcomes and operational efficiency [11]. In environmental monitoring, such approaches have been employed to forecast air quality indices and track pollution patterns, enabling timely interventions and policy adjustments [12]. By aligning our work with these related applications, we demonstrate the broader applicability and potential impact of the KAN-T-GCN model in domains beyond transportation. However, this paper will focus on transportation data to showcase the model's performance and utility in a well-defined and impactful domain.

The contributions of this paper are demonstrated through the following points:

- This paper comprehensively demonstrates the exceptional performance of Kolmogorov-Arnold Networks (KANs) in generating splines as data elements, examining their accuracy, reliability, and effectiveness in extracting meaningful and valu-

able information from raw data.

- This paper introduces an innovative KAN-T-GCN model that combines the power of Kolmogorov-Arnold Networks (KANs) with the Temporal Graph Convolutional Network (T-GCN) [13] to generate data elements enabling highly accurate traffic flow predictions.
- The paper thoroughly justifies the choice of the KAN-T-GCN model for data element generation in traffic prediction tasks, demonstrating through rigorous analysis that it achieves state-of-the-art results on real-world datasets.

II. RELATED WORK

Effectively extracting data elements has long been a challenging problem in data processing [14]. To meet the high demands for data, the data processing workflow must be both efficient and capable of producing accurate and interpretable data elements [15]. Despite the critical importance of this process, there is no widely accepted definition for data processing methods or the specific content of data elements. This section provides a brief introduction to Kolmogorov-Arnold Network (KAN) methods in machine learning, which have the potential to serve as effective data element extractors. Additionally, we use the field of traffic prediction as a case study to illustrate the application of current machine learning algorithms.

2.1 Machine Learning Techniques and Kolmogorov-Arnold Networks

Machine learning (ML) techniques are highly effective in extracting valuable information from datasets by identifying complex patterns and relationships [16]. Therefore, they could conceivably be regarded as a potential tool for constructing data elements. Multi-layer perceptrons (MLPs) are key components of machine learning models, serving as fully-connected feedforward neural networks. They are commonly used in machine learning for approximating nonlinear functions due to their ability to express complex relationships, supported by the universal approximation theorem [17]. MLPs play a foundational role in modern machine learning architectures by enabling the modeling of intricate patterns and structures in

data through multiple layers of interconnected neurons [18]. Their capacity to learn from large datasets and improve predictive accuracy has been instrumental in advancements across various domains [19]. In the context of neural networks, Kolmogorov-Arnold Networks (KANs) [9] is proposed as an alternative to Multi-layer Perceptrons (MLPs) [16]. Multi-Layer Perceptrons (MLPs) are composed of multiple layers of nodes, including input, hidden, and output layers, with each node applying a weighted sum of inputs followed by a non-linear activation function [18]. Inspired by the Kolmogorov-Arnold representation theorem, KANs differ from MLPs by placing learnable activation functions on edges instead of nodes, resulting in no linear weight matrices and using splines as parameters. Recent studies, such as [20], further underscore the potential of advanced neural architectures in enhancing spatiotemporal modeling. Their work highlights the importance of efficient parameterization and activation functions.

Kolmogorov-Arnold Networks provide significant advantages in terms of efficiency, accuracy, and interpretability. The parameters of KANs, which are splines, offer greater representational capacity compared to weight matrices. This is because curves inherently capture continuous variations and complex relationships in data more effectively than numerical values alone [21]. Therefore, KAN would require fewer parameters than MLPs, leading to smaller data element sizes and faster training times. Despite their compact size, KANs achieve comparable or superior accuracy in data fitting. Additionally, KANs feature learnable activation functions on edges, enhancing their interpretability. This makes KANs more intuitive and easier to visualize, facilitating human interaction and aiding in the next stage of data elements in the data market.

2.2 Traffic Prediction Methods

In urban computing, traffic prediction is an important research area focused on forecasting future traffic conditions using historical spatial-temporal data [22]. With the advancements in technology and the widespread use of traffic sensors, there is now access to extensive traffic databases. This availability of large datasets has enabled the development of traffic prediction. Traffic prediction tasks can be classified into

3 categories, traffic status prediction, traffic flow prediction, and travel demand prediction, playing a vital role in real-world applications such as urban congestion control, route planning, vehicle dispatching, and POI recommendation [10]. Therefore, traffic condition prediction serves as a highly practical and illustrative example of data extraction, highlighting its role as a critical factor in production processes. In this paper, we choose traffic speed prediction as an example field for simplicity.

Throughout the history of traffic data analysis, classical statistical models have always played a significant role. These models rely on historical traffic data to identify underlying patterns and make future predictions, of which the Autoregressive Integrated Moving Average (ARIMA) family of models is the most popular. Studies like those by Williams and Hoel [23] demonstrate the effectiveness of ARIMA in traffic forecasting, emphasizing its simplicity and interpretability. However, the limitations of classical models become apparent in complex, non-linear traffic scenarios where these models struggle to capture intricate dependencies and sudden fluctuations in traffic flow. Moreover, traffic prediction relies on data with both spatial and temporal dependencies, and the classical statistical models fail to capture the complex topological structure of urban road networks.

On the other hand, the rapid development of machine learning offers a more sophisticated approach to traffic prediction by leveraging their ability to learn from large datasets and model complex non-linear relationships. A recent review by [24] highlights the limitations of classical approaches like ARIMA in handling nonlinear dependencies and underscores the transformative potential of neural networks in traffic prediction, especially for spatiotemporal modeling. In the context of deep neural network architectures, Recurrent Neural Networks (RNN) are utilized to capture temporal patterns in traffic data, such as the influence of past traffic on future trends. Models such as Long Short-Term Memory (LSTM) networks [25] and Gated recurrent units (GRUs) [26], which are types of recurrent neural network (RNN), have shown significant promise in traffic forecasting. LSTMs and GRUs can handle long-term dependencies in sequential data, making them well-suited for traffic prediction tasks. Research by Ma [27] highlights the superiority of LSTMs over traditional meth-

ods, showing improved accuracy in predicting traffic flow. When RNN focus on the temporal dependency of traffic prediction, Convolutional Neural Networks (CNN) [28] excel at capturing spatial trends, like how traffic flows through road networks. Graph Convolutional Networks (GCNs) [29] pay special attention to the graph-structured data, conferring upon its parent field of CNNs. Graph Convolutional Networks (GCNs) [29] have significantly improved traffic speed prediction by effectively capturing spatial and temporal dependencies inherent in traffic data. The following review synthesizes findings from several influential studies that leverage GCNs for this purpose.

To capture the spatial and temporal dependence simultaneously, it comes naturally to devise algorithms which can facilitate both the advantages of RNN-based methods and CNN-based methods. Diffusion Convolutional Recurrent Neural Network (DCRNN) [30] uses bidirectional random walks to capture spatial dependencies and an encoder-decoder architecture with scheduled sampling for temporal dependencies. Spatial-Temporal Graph Convolutional Networks(ST-GCN) [31] applies GCNs to point-of-interest (POI) recommendation, addressing the spatial and temporal dependencies in user preferences. Multivariate Time-Series Graph Neural Network(MTGNN) [32] focuses on multivariate time-series forecasting using graph neural networks to capture dependencies among multiple traffic-related variables. Traffic Graph Convolutional Long Short-Term Memory(TGC-LSTM) [33] combines GCN with LSTM, whereas Traffic Graph Convolutional Networks(T-GCN) [13] integrates GCNs with GRUs to model spatial-temporal dependencies in traffic data. In recent years, the development of spatial-temporal large language models, such as ST-LLM [34] and TPLLM [35], has revolutionized traffic prediction by effectively capturing complex dependencies in traffic data, leading to more accurate and reliable forecasts. These algorithms have significantly improved traffic speed prediction by effectively capturing spatial and temporal dependencies inherent in traffic data.

III. METHODOLOGY

3.1 Problem Formulation

The underlying property of the traffic speed prediction

process is just the prediction of future traffic speed based on past traffic information. When we say past traffic information, it may include not only past traffic speeds but also external information including road characteristics, traffic control measures, weather conditions, incidents, roadworks and so on. However, it would be arduous or even impossible to collect all these information. Therefore, it is beyond reproach to chiefly focus on the past traffic speeds and the most accessible external information, the road network structure.

3.2 Mathematical Representation of Data

In the context of traffic forecasting, $G = (V, E)$ represents the road network as a graph, where each road segment is a node represented by V , and E are the edges. The connections between roads are illustrated by an adjacency matrix A . Given a total number N of nodes, we define $V = v_1, v_2, \dots, v_N$, where v_i is the i -th node. The adjacency matrix $A \in \{0, 1\}^{N \times N}$ consists of 0s and 1s, with 0 indicating no connection between roads and 1 indicating a connection. This method helps to comprehend the topological structure of the road network and the interconnections between roads for predicting traffic patterns. Given the number of the historical time points T , the feature matrix $\mathbf{X} \in \mathbb{R}^{N \times T}$ encapsulates the traffic information attributes for each node in the road network, specifically traffic speed in our case. Each element $\mathbf{X}_{i,j}$ in the feature matrix denotes the traffic speed on node i at time j , where $i \in \{1, \dots, N\}, j \in \{t - T + 1, \dots, t - 1, t\}$. Thus, each column of the feature matrix, written as $\mathbf{X}_j$, represents the traffic speed on the road network G at time j . This matrix is crucial for the spatio-temporal traffic forecasting tasks, enabling the model to learn the relationship between the road network's topology and historical traffic data attributes for prediction purposes.

3.3 KAN-T-GCN Model Architecture

As shown in Figure 1, the KAN-T-GCN model consists of three parts: KAN-based Graph Convolutional Network (GCN), Gated Recurrent Unit (GRU) and a KAN layer. Each of these three components is responsible for distinct aspects of the traffic speed dataset: spatial dependence, temporal dependence, and output formatting, respectively. The KAN-based GCN

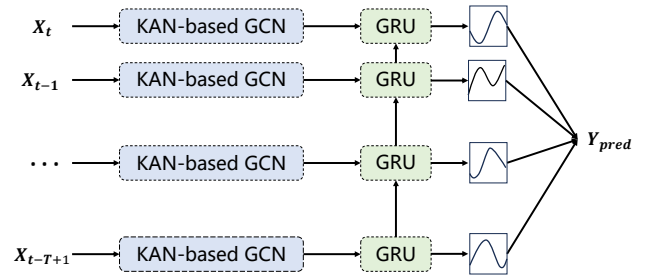


Figure 1. The KAN-T-GCN Model Architecture, consisting of three primary components: (1) a KAN-based Graph Convolutional Network (GCN) for spatial dependency extraction, (2) a Gated Recurrent Unit (GRU) for temporal dependency modeling, and (3) an output KAN layer for refining and generating the final predictions.

component forms the initial part of the model. It processes sequential inputs $(\mathbf{X}_{t-T+1}, \dots, \mathbf{X}_{t-1}, \mathbf{X}_t)$, where T represents the time window size. Each $\mathbf{X}_j$ (where $t - T + 1 \leq j \leq t$) is a data point at time j . The KAN-based GCN consists of multiple layers, each of which captures both graph-structured data and domain-specific knowledge. These layers are stacked to refine the feature representation progressively. By leveraging learnable spline-based activation functions, the KAN-based GCN efficiently models spatial relationships while reducing parameter complexity and maintaining high accuracy. The second component of the model is the GRU network, which processes the sequential output from the KAN-based GCN. GRUs excel in capturing temporal dependencies in sequential data by using update and reset gates to manage information flow. This reduces computational overhead compared to traditional RNNs and mitigates vanishing gradient issues, making GRUs particularly well-suited for long-term traffic forecasting. The final component of the model is an additional KAN layer, which processes the output of the GRU. This layer integrates domain knowledge once more, refining the temporal features extracted by the GRU and output the the model's final prediction of future traffic speed Y_{pred} based on the processed inputs for a desired length of time.

3.3.1 The KAN-Based Graph Convolutional Network

In this section, we will carefully show how to construct the KAN-based GCN component mathematically. A layer of the famous Graph Convolutional

Network(GCN) can be mainly decomposed into three parts: normalization, aggregation and activation. Directly multiplying the adjacency matrix A with the feature matrix $\mathbf{X}$ without normalization would result in nodes with a higher degree having disproportionately large feature values since they aggregate features from more neighbours. This can cause numerical instability and bias in the learning process. The normalization step is inspired by the normalized graph Laplacian. It aims to ensure that the feature aggregation is normalized by the node degrees, which equalizes the influence of each neighbour. The normalized adjacency matrix $\hat{A}$ with degree $N \times N$ is given by:

$$\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}, \quad (1)$$

where:

- $\tilde{A} = A + \mathbf{I}_N$ is the adjacency matrix with added self-loops,
- $\tilde{D}$ is the degree matrix of $\tilde{A}$, defined as $\tilde{D}_{ii} = \sum_{j=1}^N \tilde{A}_{ij}$ for $1 \leq i \leq N$,
- $\tilde{D}^{-1/2}$ is the inverse square root of the degree matrix $\tilde{D}$.

The aggregation step in a GCN is designed to combine the features of a node's neighbors to form a new feature representation for the node itself. This step captures the local structure of the graph and allows the network to learn from the connectivity patterns of the nodes. Therefore, the new feature representation is written by:

$$\mathbf{Z}^{(l+1)} = \hat{A} \mathbf{H}^{(l)} \mathbf{W}^{(l)}, \quad (2)$$

where:

- $\hat{A}$ is the normalized adjacency,
- $\mathbf{H}^{(l)}$ is the feature matrix at layer l with $\mathbf{H}^{(0)} = \mathbf{X}$,
- $\mathbf{W}^{(l)}$ is the weight matrix at layer l .

The primary purpose of the activation function σ is to introduce non-linearity into the model. Linear models can only learn linear relationships, which severely limits their ability to capture the complexity of real-world data. By applying a non-linear activation function, the model can learn complex, hierarchical representations of the data. Therefore, a typical GCN layer can be expressed as

$$\mathbf{H}^{(l+1)} = \sigma(\mathbf{Z}^{(l+1)}), \quad (3)$$

with σ being the activation function at layer l .

In our KAN-T-GCN model, the aggregation and activation steps are merged following the KAN model's approach, utilizing a learnable activation function followed by a summation.

In the concept of the KAN model, the concept of residual activation functions involves incorporating a basis function, denoted as $b(x)$, into the activation function $\phi(x)$:

$$\phi(x) = w_b b(x) + w_s \text{spline}(x), \quad (4)$$

where:

- w_b and w_s are weights kept for fine-tuning purposes applied to the basis function and splines component respectively,
- $b(x)$ is the basis function, defined as $b(x) = \text{silu}(x) = \frac{x}{1+e^{-x}}$,
- $\text{spline}(x)$ is a linear combination of basis splines $B(x)$, defined as $\text{spline}(x) = \sum_{i=0}^{k+G-1} c_i B_i(x)$ given the piecewise polynomial order of the splines k and number of intervals G , Figure 2 illustrates an example of a spline function with order $k = 3$ as a linear summation of B-splines over $G = 5$ intervals.

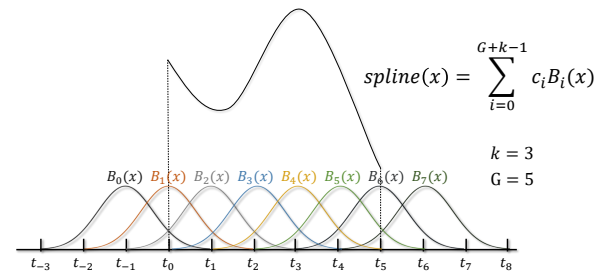


Figure 2. $\text{Spline}(x)$.

By adding the basis function $b(x)$ to the spline function within the activation function $\phi(x)$, the network can better control the magnitude and behaviour of the activation, enhancing the training process, where c_i are coefficients that can be adjusted during training. We then write a matrix $\Phi^{(l)}$ with dimensions $(n_{l+1} \times n_l)$ contains the activation functions $\phi_{l,j,i}(\cdot)$ that connects neuron (l, i) and neuron $(l + 1, j)$. Therefore, the aggregation step and the activation step of the KAN-T-GCN model responsible for the spatial dependence

can be formulated as:

$$\mathbf{Z}^{(l+1)} = \hat{A}\mathbf{H}^{(l)}, \quad (5)$$

$$\mathbf{H}^{(l+1)} = \Phi^{(l)}\mathbf{Z}^{(l+1)}, \quad (6)$$

where $\Phi^{(l)}$ is the matrix of activation functions at layer l .

In conclusion, we employ a KAN-modified GCN model to extract spatial features from traffic speed data. For simplicity, a two-layer KAN-modified GCN model is used for our model and it is written as:

$$\hat{X} = f(\mathbf{X}, A) = \mathbf{H}^{(2)} = \Phi^{(1)}(\hat{A}(\Phi^{(0)}(\hat{A}\mathbf{X}))). \quad (7)$$

A two-layer KAN-modified GCN model with only two input x_t and 5 hidden neurons is presented in Figure 3. This figure emphasizes the mechanics of the KAN-based GCN. Each layer applies normalization to the adjacency matrix to stabilize feature aggregation and prevent numerical biases from high-degree nodes. The spline-based activation functions integrate local and global spatial features, providing enhanced interpretability and efficiency compared to traditional weight matrices. This refined spatial data is then passed to the temporal modeling component (GRU).

3.3.2 the Gated Recurrent Unit

The output of the above equation will be fed to the gated recurrent units (GRUs) shown in Figure 4. The equations involved are:

$$u_t = \sigma(W_u[\hat{X}_t, h_{t-1}] + b_u), \quad (8)$$

$$r_t = \sigma(W_r[\hat{X}_t, h_{t-1}] + b_r), \quad (9)$$

$$c_t = \tanh(W_c[\hat{X}_t, (r_t * h_{t-1})] + b_c), \quad (10)$$

$$h_t = u_t * h_{t-1} + (1 - u_t) * c_t, \quad (11)$$

where:

- u_t is the update gate at time t ,
- r_t is the reset gate at time t ,
- c_t is the candidate hidden state at time t ,
- h_t is the new hidden state at time t .

The Gated Recurrent Unit (GRU) is designed to address the vanishing gradient problem in recurrent neural networks by incorporating two gates: the reset gate r_t and the update gate u_t . The reset gate determines how much of the past information is ignored when computing the candidate hidden state c_t . If r_t is close to 0, the model discards most of the previous information, allowing it to focus on recent data. The update gate u_t , on the other hand, controls how much of the previous hidden state h_{t-1} is carried forward to the current state h_t . When u_t is close to 1, the GRU retains more past information, facilitating long-term dependencies. Together, these gates enable the GRU to dynamically balance between remembering and forgetting, which is crucial for modeling temporal dependencies in traffic data.

3.3.3 Output Formatting

For a classic GRU network, a fully connected layer is used afterwards to transform the high-dimensional hidden state vectors h_t produced by the GRU into a suitable format for the final task. This MLP layer combines and refines the features learned by the GRU, applying a linear transformation followed by a non-linear activation function to model complex relationships. For our KAN-T-GCN model, this fully connected layer is replaced by a KAN layer

$$Y_{pred} = \Phi h, \quad (12)$$

where Φ is the matrix of activation functions similar to Φ in Equation (6). The matrix $\Phi^{(l)}$ in Equation (6) represents the collection of activation functions applied to each connection between nodes in layer l . Each element of $\Phi^{(l)}$ is a learnable activation function $\phi_{l,j,i}(x)$, constructed as a combination of a basis function $b(x)$ and spline functions as described in Equation (5). The basis function introduces non-linearity, while the spline functions allow for fine-grained control over the activation behavior. The coefficients w_b and w_s determine the contributions of the basis and spline components, respectively. This construction ensures flexibility and adaptability of the activation functions to the underlying data distribution.

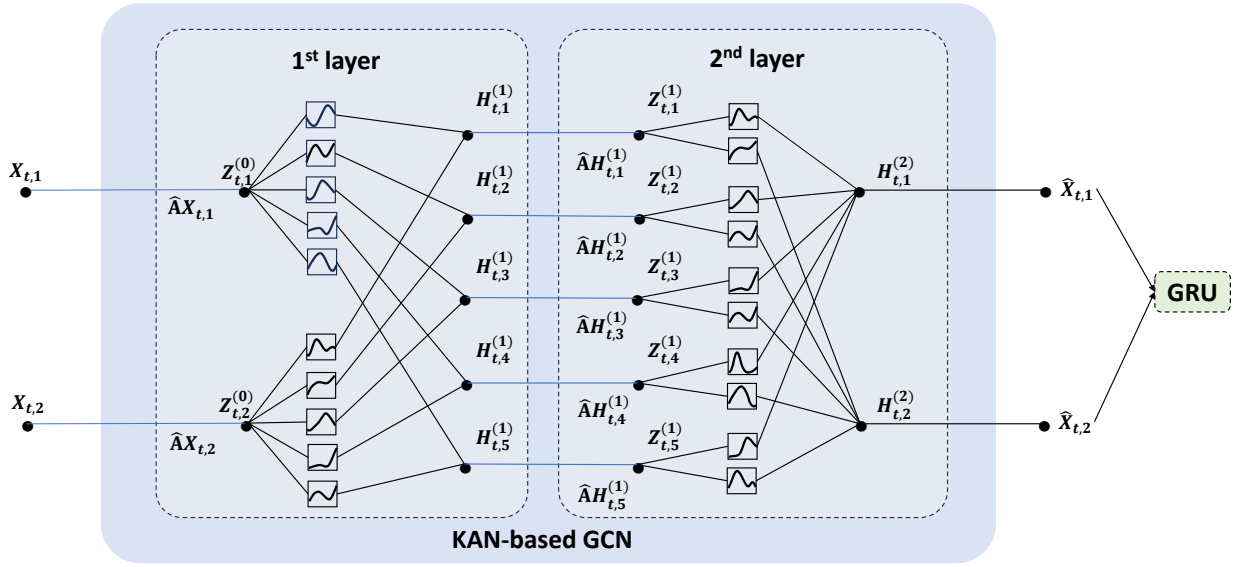


Figure 3. 2-layer KAN-based GCN architecture with 2 input features, 2 output features and 5 hidden neurons.

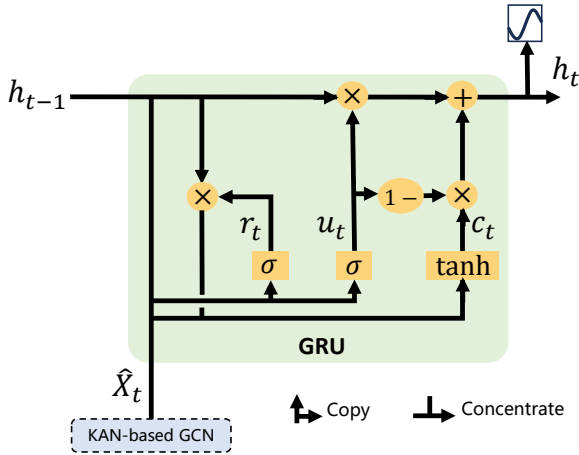


Figure 4. The gated recurrent unit.

3.3.4 Loss Function

In the training phase of the KAN-T-GCN model, the objective is to reduce the difference between the actual traffic speed on roads Y and the predicted speed Y_{pred} . This is achieved by minimizing the loss function, which consists of two terms: the first term minimizes the error between the actual and predicted speeds, while the second term, L2 regularization (L_{reg}), helps prevent overfitting by adjusting the model complexity using a hyperparameter λ .

$$\text{loss} = \|Y - Y_{pred}\| + \lambda L_{reg}. \quad (13)$$

IV. EXPERIMENTS

4.1 Dataset and Experimental Setup

4.1.1 Dataset

In this section, we employ three real-world datasets [36] pertaining to traffic speeds in Shenzhen, Los Angeles, and the San Francisco Bay Area. These datasets are referred to as “SZ_Taxi”, “METR_LA”, and “PEMS_BAY” respectively. Each dataset consists of an adjacency matrix detailing spatial relationships between roads and a feature matrix showing speed variations over time for each road with a road ID.

SZ.Taxi comprises traffic speeds of taxis in Shenzhen from January 1 to January 31, 2015. The dataset focuses on 156 primary roads within the Luohu District, with data aggregated at 15-minute intervals, resulting in 464,256 speed records. Cleaning steps included removing records with implausible speeds and interpolating missing values using linear interpolation. METR.LA includes traffic speed data collected from sensors embedded in the highway across Los Angeles County. This dataset encompasses data from 207 sensors over a four-month period, from March to June 2012, with recordings taken every 5 minutes. This results in a total of 34,272 time slices and 7,094,304 speed records. Data was collected by the Los Angeles Metropolitan Transportation Authority using loop de-

tectors embedded in road pavements. The limitations include potential inaccuracies caused by hardware malfunctions or environmental factors affecting sensor readings. Preprocessing involved identifying faulty sensor readings through anomaly detection algorithms and imputing missing data using spatiotemporal interpolation. PEMS_BAY consists of traffic speed data obtained from 325 sensors deployed throughout the Bay Area. This data is collected by the California Transportation Agencies (CalTrans) Performance Measurement System (PEMS). The dataset spans a six-month period, from January to May 2017, with information recorded at 5-minute intervals, amounting to 52,116 time slices and 16,937,700 speed records. For all datasets, the preprocessing aimed to standardize and normalize the traffic speed values to ensure consistency across different datasets. Additionally, we acknowledge limitations such as biases in spatial coverage and temporal coverage gaps caused by external factors such as road construction or weather conditions. In the experiment, we divide each dataset into three non-overlapping sets. The earliest 70% of the data is used as the training set, the subsequent 20% as the validation set, and the final 10% as the test set.

4.1.2 Evaluation Metrics

Given the number of traffic speed prediction tasks n , we employ three metrics to evaluate the difference between the prediction result $Y_{pred}(i)$ and the real traffic speed $Y(i)$ of task i including:

1) Mean Square Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_{pred}(i) - Y(i))^2. \quad (14)$$

2) Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|Y_{pred}(i) - Y(i)|}{Y(i)}. \quad (15)$$

3) Coefficient of determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y(i) - Y_{pred}(i))^2}{\sum_{i=1}^n (Y(i) - \bar{Y})^2}, \quad (16)$$

where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y(i)$.

Specifically, for MSE and MAPE, lower values indicate better performance, whereas for R^2 , higher values indicate better performance. For all prediction tasks, we set the time window T for input and the time window T' for the target to 12.

4.1.3 Comparison Setup

Since the KAN-T-GCN model is derived from the T-GCN model by combining GCN and GRU networks, we selected the well-established T-GCN and GCN as the first two base models for our performance comparison. In addition, we select the ST-GCN [31] model as our third base model, which also addresses both spatial and temporal dependencies, but pays special attention to user preferences. In order to fairly evaluate the performance difference, we ensure that the spatial components of KAN-T-GCN, T-GCN, and ST-GCN models have an equal number of layers and a similar number of trainable parameters. The weight of GCN, T-GCN, and ST-GCN models are initialized according to the Xavier initialization. In the KAN-T-GCN model, each activation function as shown in equation (4) is initialized to have $w_s = 1$ and $c_i \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.1$ follow closely to the original KAN model [9].

We set the batch size to 128 and trained the model for 300 epochs using an early-stopping strategy. All evaluations were conducted on a system running Ubuntu 18.04.6 LTS, equipped with an Intel(R) Xeon(R) Gold 6248 CPU, a Tesla V100 GPU, and 128 GB of memory.

4.2 Experimental Results and Discussion

4.2.1 Overall Performance

For simplicity, we set the number of layers for the spatial components of KAN-T-GCN, GCN, T-GCN, and ST-GCN models to 2, with the number of hidden neurons in each layer in spatial components denoted as h_{KAN} , h_{GCN} , h_{T-GCN} , and h_{ST-GCN} , respectively. Given the complexity and size of the real datasets, we choose relatively large values for h_{GCN} , h_{T-GCN} , and h_{ST-GCN} , specifically 256. For the KAN-T-GCN model, we additionally define the piecewise polynomial order of the splines, k , and the number of intervals, G to ensure that KAN-T-GCN, GCN, T-GCN, and ST-GCN models share a similar number of trainable parameters in the hidden layers. For MLP-

based GCNs including GCN itself, T-GCN, and ST-GCN models, the number of parameters can be calculated as $h \times h$, assuming the hidden layers have the same number h of hidden neurons. For the KAN-T-GCN model, the mainly trainable parameters are c_i s in $spline(x)$ as defined in equation (4), where w_b and w_s are absorbed into $b(x)$ and $spline(x)$ in our experiment. Therefore the number of parameters of a hidden layer of the KAN-T-GCN model is the number of c_i s, which is calculated as $(G + k - 1) \times h_{KAN}$. Cubic splines are demonstrated to be commonly good for KAN-structured networks [9]. We thus choose the activation functions to be common cubic splines with $k = 3$ and the number of intervals $G = 5$, and thus h_{KAN} is calculated to be 36.

The experimental results presented in Table 1 highlight the superiority of our model, KAN-T-GCN, over other models (GCN, ST-GCN, T-GCN) across three datasets: SZ-TAXI, PEMS-BAY, and METR-LA. Our model consistently achieves the lowest Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) scores, demonstrating its enhanced predictive accuracy.

For instance, in the SZ-TAXI dataset, KAN-T-GCN achieves an MSE of 14.7823 and a MAPE of 0.0124, which are significantly lower compared to the best-performing competing model, STGCN, which has an MSE of 15.2579 and a MAPE of 0.0242. This represents a reduction in MSE by approximately 3.12% and in MAPE by 48.76%. Similarly, in the PEMS-BAY dataset, KAN-T-GCN achieves an MSE of 14.4153 and a MAPE of 0.0402, outperforming the closest competitor, STGCN, with an MSE of 14.4828 and a MAPE of 0.0356. This shows a minor improvement in MSE but a significant enhancement in MAPE, showcasing the reliability of our model in reducing prediction errors.

Moreover, KAN-T-GCN achieves the highest R-squared (R^2) values across all datasets, indicating a better fit and more accurate predictions. For the SZ-TAXI dataset, KAN-T-GCN achieves an R^2 value of 0.8619, compared to the best competing model, TGCN, with an R^2 of 0.8404. For the PEMS-BAY dataset, KAN-T-GCN achieves an R^2 of 0.8333, surpassing TGCN's R^2 of 0.8226. In the METR-LA dataset, while KAN-T-GCN's R^2 value of 0.6849 is slightly lower than GCN's 0.6857, it still provides a competitive fit with significantly better MSE and

MAPE scores.

The integration of Kolmogorov-Arnold Networks (KAN) into the Temporal Graph Convolutional Network (T-GCN) framework significantly enhances the model's ability to capture complex nonlinear relationships and temporal dynamics within the traffic network. KANs are based on the Kolmogorov-Arnold representation theorem, which states that any multivariate function can be represented as a superposition of functions of one variable. The core advantage of incorporating Kolmogorov-Arnold Networks (KAN) into the T-GCN framework lies in KAN's ability to model highly nonlinear functions. In the context of traffic speed prediction, this means that KANs can approximate the complex relationships between traffic speeds, road conditions, and other factors that influence traffic flow. This enhanced representation learning capability allows KAN-T-GCN to produce predictions with lower Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE), indicating a closer match between the predicted and actual traffic flows.

The R^2 value, a metric of goodness-of-fit, assesses how well the model explains the variability in the observed data. The superior R^2 scores achieved by KAN-T-GCN suggest that the model not only predicts accurately but also does so in a manner that closely follows the underlying trends and patterns in the traffic data. By replacing the traditional neurons in T-GCN with KAN units, the model gains the capability to learn and represent intricate, nonlinear interactions between the various components of the traffic system. This can be attributed to KAN's ability to model higher-order interactions and temporal dependencies in a more comprehensive manner compared to the baseline models. As a result, KAN-TGCN offers a more interpretable representation of the traffic system dynamics, enhancing its R^2 value.

Traditional GCNs excel at capturing spatial dependencies in graph-structured data, but they lack the temporal modeling capabilities required for traffic flow prediction. GCNs operate by aggregating information from neighboring nodes in the graph, updating each node's representation based on its local neighbourhood. While this approach is effective for tasks involving static graphs, it tends to underperform in tasks requiring predictions of future states when applied to dynamic systems like traffic networks, where node rep-

Table 1. Overall comparison.

Model	SZ_TAXI			PEMS_BAY			METR_LA		
	MSE	MAPE	R^2	MSE	MAPE	R^2	MSE	MAPE	R^2
GCN	18.3972	0.0315	0.8307	23.7210	0.0573	0.7444	162.2813	0.1385	0.6857
ST-GCN	15.2579	0.0242	0.7395	14.4828	0.0356	0.7239	131.0972	0.1106	0.6261
T-GCN	17.3385	0.0275	0.8404	16.4577	0.0405	0.8226	148.9741	0.1257	0.7115
KAN-T-GCN	14.7823	0.0124	0.8619	14.4153	0.0402	0.8333	99.0578	0.1004	0.6849

representations and relationships evolve over time.

ST-GCN and KAN-T-GCN address this limitation by incorporating temporal modelling mechanisms such as GRU alongside the spatial modelling capabilities of GCNs. However, KAN-T-GCN goes a step further by enhancing the spatial modelling with KANs, which are better suited to capturing the complex, nonlinear interactions that arise within traffic networks. In KAN-T-GCN, the spatial and temporal modelling components work synergistically to produce accurate traffic flow predictions. The KAN units capture the nonlinear spatial interactions between nodes in the traffic network, while the temporal modelling mechanisms (e.g., RNNs or CNNs) capture the temporal evolution of these interactions over time. This interplay between spatial and temporal modelling enables KAN-TGCN to generate predictions that are both accurate and highly interpretable. The accuracy comes from the model's ability to represent the full range of nonlinear dynamics present in the traffic system, while the interpretability arises from the clear separation of spatial and temporal modeling components within the model architecture.

In summary, the enhanced performance of KAN-TGCN in traffic flow prediction can be attributed to its ability to capture complex nonlinear interactions both spatially and temporally within the traffic network. By combining the strengths of KANs, GCNs, and temporal modeling mechanisms, KAN-TGCN offers a more comprehensive and accurate representation of traffic dynamics, resulting in predictions with lower MSE, MAPE, and higher R^2 values compared to traditional GCNs and even ST-GCNs.

4.2.2 Ablation Test

In the context of Kolmogorov–Arnold Networks (KANs), the number of grid intervals (denoted as G) can significantly affect the network's performance. As demonstrated in the original KAN paper [9], starting with G values between 3 and 10 typically yields good

performance. In practice, Kiamari [37] found that this range of G provides a good ability to model complex graphic data, reflecting that this range is empirically found to be effective in many cases. Therefore, in our KAN-T-GCN model, we explore the impact of G by selecting $G = \{5, 6, 7, 8, 9, 10\}$ to build upon these findings and ensure robust results. Increasing G improves the resolution of the grid used to approximate the function. More grid intervals mean that the network can more accurately capture the nuances of the function, which means the choice of G affects how well the KAN can approximate the function. However, this also increases the computational complexity and may require more data to achieve reliable results. Choosing $G = 5$ to $G = 10$ offers a practical trade-off between accuracy and computational efficiency, making it manageable to train and validate the network. Thus, we strike a balance between having a grid that is fine enough to capture complex patterns but not so fine that the network becomes unwieldy or prone to overfitting.

MAPE and MSE both provide a measure of how well the model is performing, with lower values indicating better performance. MSE and MAPE calculate the average of the squared differences and average absolute percentage error between predicted and actual values respectively. As a result, MSE emphasizes larger errors more heavily due to squaring the errors, penalizing larger errors more severely, whereas MAPE provides a percentage-based measure, making it scale-independent and easier to interpret in terms of relative error. In the case of traffic speed prediction, it is more intuitive to choose MAPE in order to understand the error in terms of the relative size of the errors compared to the actual values. Thus, in the ablation test section, we focus on MAPE and R^2 errors.

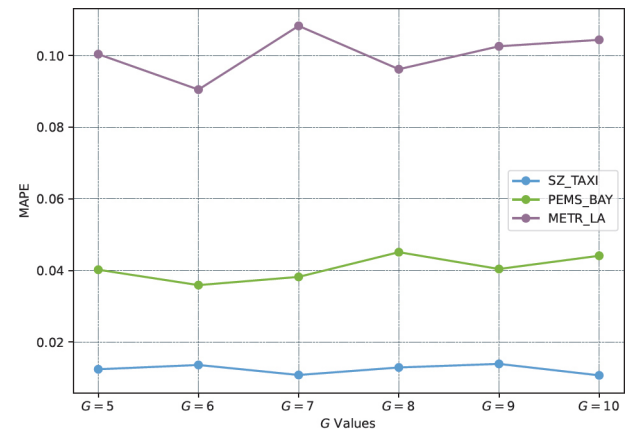
As shown in Figure 5a, MAPE for the three datasets SZ_TAXI, PEMS_BAY, and METR_LA indicate strong performance across various G values. For the SZ_TAXI dataset, the MAPE values are consis-

tently low, ranging between 0.0108 and 0.0139. Similarly, the PEMS_BAY dataset shows MAPE values between 0.0359 and 0.0451, reflecting good predictive accuracy. The METR_LA dataset, despite having slightly higher MAPE values ranging from 0.0905 to 0.1083, still demonstrates reasonably good performance. Part(a) illustrates that the performance of the KAN-T-GCN model measured by MAPE remains stable and consistent regardless of the changes in the G parameter. The architecture of KAN-T-GCN inherently supports robustness and generalizability. By efficiently capturing the underlying data structure with fewer parameters and flexible functions, KAN-T-GCN maintains stable performance even when the G parameter varies. This robustness ensures that the model is not overly sensitive to the choice of G , leading to consistent MAPE values.

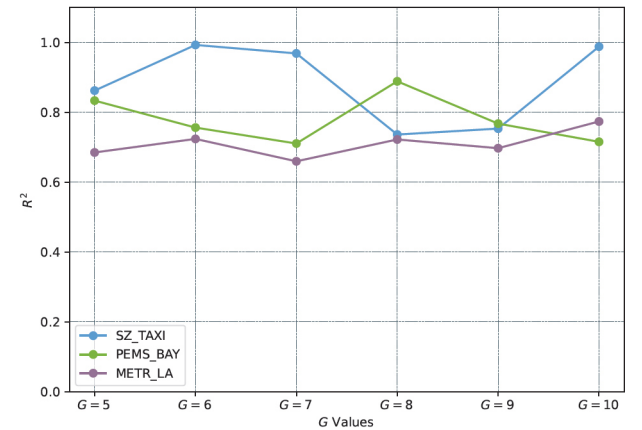
In Figure 5b, the R^2 values for the three datasets are consistently high, providing further evidence of good model performance. The R^2 values for SZ_TAXI are particularly strong, often nearing 1.0, which signifies that the model explains nearly all the variability in the data. The METR_LA dataset shows the lowest R^2 values among the three but still maintains values around 0.65 to 0.75, which are indicative of a strong model fit. While the MAPE values showed stability across different G values, it is evident that the R^2 values exhibit more fluctuation, suggesting that R^2 is more sensitive to changes in the G parameter. For instance, the R^2 values for SZ_TAXI vary significantly, with some dips around $G = 8$ and $G = 9$, and peaks at $G = 6$ and $G = 10$. This fluctuation indicates that while the model maintains a good fit, its explanatory power varies more noticeably with different G values. This sensitivity is due to the model's capacity to capture the variance in the data, which is more influenced by the structural changes in G compared to the absolute errors captured by MAPE.

In conclusion, the KAN-T-GCN model demonstrates robust performance across different G values for the three datasets, SZ_TAXI, PEMS_BAY, and METR_LA. The MAPE values remain consistently low, indicating stable and reliable predictive accuracy. On the other hand, R^2 values, while generally high, show more sensitivity to changes in G , highlighting the importance of considering R^2 when selecting the appropriate G value. For optimal performance, the best G values observed are $G = 10$ for SZ_TAXI,

$G = 7$ for PEMS_BAY, and $G = 6$ for METR_LA, as these values yield the highest R^2 scores, reflecting the model's ability to explain the variability in the data effectively. While varying G leads to distinct changes in model accuracy, it also directly influences the learned function in an interpretable manner. This contrasts with traditional MLPs, where hyperparameter adjustments primarily affect optimization dynamics rather than the model's inherent transformations. This further supports the claim that the internal mechanisms of the KAN-T-GCN model can be analyzed through its architecture.



(a) MAPE for different G values.



(b) R^2 for different G values.

Figure 5. Comparison of MAPE and R^2 for different G values.

V. CONCLUSION

This study presents a significant advancement in the field of data processing by introducing the KAN-

T-GCN model, which effectively generates interpretable data elements from raw data. The research underscores the importance of converting raw data into actionable insights, especially in the context of the rapidly evolving data economy. By integrating Kolmogorov-Arnold Networks (KAN) with Temporal Graph Convolutional Networks (T-GCN), the KAN-T-GCN model addresses the critical need for precision, dynamic update capability, and interpretability in data elements.

The model was rigorously tested using three comprehensive real-world traffic datasets: SZ_Taxi from Shenzhen, METR_LA from Los Angeles, and PEMS_BAY from the San Francisco Bay Area. The results demonstrated that KAN-T-GCN consistently achieved lower Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) compared to traditional models, while also providing higher R-squared (R^2) values, indicating a better fit to the observed data. These improvements can be attributed to the model's enhanced ability to capture complex nonlinear spatio-temporal dependencies inherent in traffic data.

One of the key contributions of this work is the detailed exploration of the KAN-based Graph Convolutional Network (GCN) and its integration with Gated Recurrent Units (GRUs) to form the KAN-T-GCN model. This novel architecture synergizes the spatial modeling strengths of KAN-based GCNs with the temporal modeling capabilities of GRUs, enabling precise and interpretable predictions. The innovative use of Kolmogorov-Arnold splines as activation functions further enhances the model's expressiveness and efficiency, reducing parameter complexity while maintaining high accuracy. Additionally, the model's capacity to update dynamically with new data ensures that it remains relevant and accurate over time, addressing one of the major challenges in data processing. The ablation tests conducted to examine the impact of different grid interval values G further validated the robustness and flexibility of the KAN-T-GCN model. The results indicated that while Mean Absolute Percentage Error (MAPE) values remained stable, R^2 values showed sensitivity to changes in G , suggesting optimal configurations for different datasets.

The advantages of the KAN-T-GCN model over existing approaches are manifold: its ability to provide interpretable data representations, its superior predic-

tive performance, and its robustness across diverse datasets and configurations position it as a transformative tool in data-driven decision-making. In conclusion, the KAN-T-GCN model represents a robust and innovative solution for generating interpretable data elements from raw data, with proven efficacy in traffic prediction tasks. The model's superior performance in handling spatio-temporal data makes it a valuable tool for enhancing data-driven decision-making processes.

Future research will focus on several key areas to extend the capabilities and applications of the KAN-T-GCN model. First, the model will be evaluated across different domains beyond transportation, such as healthcare, finance, and environmental monitoring, to assess its generalizability and effectiveness in various contexts. According to specific real-world scenarios, we will explore a more comprehensive hyperparameter tuning process, including the number of hidden layers, learning rates, and other model-specific parameters, using advanced techniques such as Bayesian optimization. This evaluation will help identify domain-specific challenges and validate the model's adaptability for diverse spatiotemporal prediction tasks. Second, we will explore the impact of changing other parameters within the model, such as the number of hidden layers, the type of activation functions, and the learning rate, to optimize performance further. Additionally, we will experiment with advanced optimization techniques such as Bayesian optimization and adaptive learning rate schedulers to systematically tune hyperparameters and improve model performance across varied datasets. Since the KAN-T-GCN model inherently has interpretability due to its spline-based structure, we will further explore whether techniques such as Layer-wise Relevance Propagation (LRP) and SHAP can enhance its explainability by providing deeper insights into the learned functions within the KAN component. While this study primarily focuses on evaluating KAN-T-GCN against structurally similar spatiotemporal models, future research will explore extending KAN's applicability to Transformer-based architectures. Specifically, we aim to investigate whether replacing MLP layers in self-attention mechanisms with KAN-based components can improve model performance while preserving interpretability. Third, we will consider the visualization of the data elements generated by our model, providing intuitive and accessible insights for

end-users. Interactive dashboards and visualization tools will be developed to enable stakeholders to explore and interpret patterns and relationships extracted by the model, enhancing its usability for decision-making processes. Finally, we will investigate how to integrate the curve-based data elements into the broader data industry, focusing on their practical applications and potential to enhance data sharing and utilization across sectors. This will include targeted use cases such as dynamic traffic management systems, predictive healthcare platforms, and financial analytics tools, with a focus on collaborating with industry stakeholders to ensure real-world impact.

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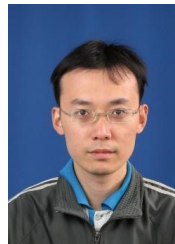
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